AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2015

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

i. Find the value of x if $5:2x = 3:2x - 4$. (2 Marks)

ii. The number of students in a school increases by 25% each year. If there were 2,400 students enrolled in the year 2014, find the enrolment for the previous year 2013. (1 Mark)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. After her death, Ms Salma's property of Rs 100,000 was divided among her husband, mother and daughter in the ratio 3 : 2 : *x*. Find the value of *x* if the daughter's share is Rs 50,000.

b. Kulsoom has gold jewellery weighing 500 grams and silver jewellery worth Rs 8,600. Find the amount of Zakat she has to pay if the price of gold is Rs 40,500 for 10 grams.

[Note: Rate of Zakat is 2.5% of annual saving.]

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a. At a pizza restaurant, the prices of large and medium size pizza are Rs 1,650 and Rs 1,050 respectively. They offer the following discount offers.

DISCOUNT OFFER 1:	BUY 2 LARGE SIZE PIZZAS, GET 15% OFF
DISCOUNT OFFER 2:	BUY 1 LARGE AND 1 MEDIUM SIZE PIZZA, GET 5% OFF

If a family buys 1 medium and 3 large size pizzas from this restaurant, and they avail both discount offers, what is the amount of their bill?

b. A man sells two camels for Rs 140,000 EACH. If he sells one camel at a profit of 25% and the other camel at a loss of 20%, find the profit or loss the man made in this deal.

Q.4. (Total 3 Marks)

Noor leased a car worth Rs 2,500,000 with a down payment of 65%. If the rate of interest on the remaining amount is 5% for a period of 2 years, find the total interest Noor would have paid for this car by the end of 2 years.

Q.5. (Total 5 Marks)

The salary schedule for a sales agent of a pharmaceutical company is as follows:

Weekly fixed pay for a sales agent	Rs 3,000
Overtime per hour	Rs 300
Commission on every order less than Rs 5,000	3% of sale
Commission on every order greater than Rs 5,000	5% of sale

In a certain week, this sales agent works 4 extra hours and he books orders of Rs 2,000 and Rs 11,000. What will be his salary of this week?

[Note: Commission is calculated for every order separately.]

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a.

i. Find the value of x if $\frac{1}{2}\log_4(3x-2)=\frac{1}{2}$. (3 Marks)

ii. Simplify $\frac{\log_3 9}{\log_3 3}$. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b.

i. Simplify $\sqrt{\frac{128m^{-3}n^3}{2m^{-7}n^9}}$, giving your answer in **positive** exponents. (4 Marks)

ii. Convert 7.6×10^9 from scientific form to ordinary form. (1 Mark)

Q.7. (Total 4 Marks)

The nth term of a sequence is $a_n = 5 - 3n$, where $n = 1, 2, 3\ldots$ Prove that it is an arithmetic sequence.

(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 5 Marks)

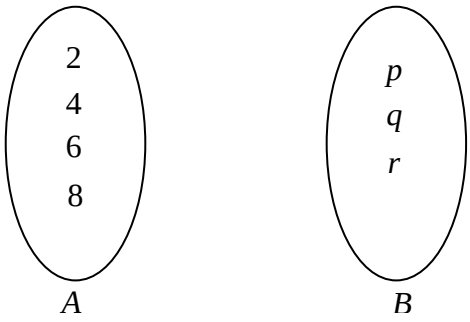
a.

i.

I. Find domain and range of the binary relation $r = \{(1,0),(1,1),(1,2),(2,0),(2,1),(2,2)\}$. (2 Marks)

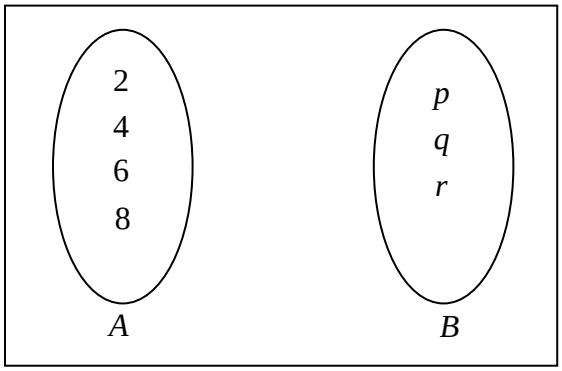
II. Is $r = \{(1,0),(1,1),(1,2),(2,0),(2,1),(2,2)\}$ a function? Write a statement to justify your answer. (1 Mark)

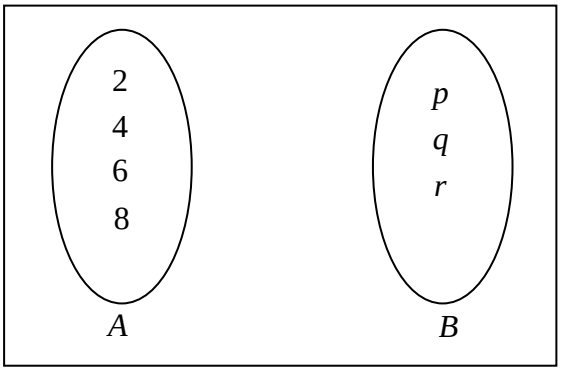
ii. Two sets *A* and *B* defined as $A = \{2,4,6,8\}$ and $B = \{p,q,r\}$ are shown below.



Join the elements of set *A* with the elements of set *B* to illustrate

I. an **into** function from *A* to *B*. II. an **onto** function from *A* to *B*. (2 Marks)





b.

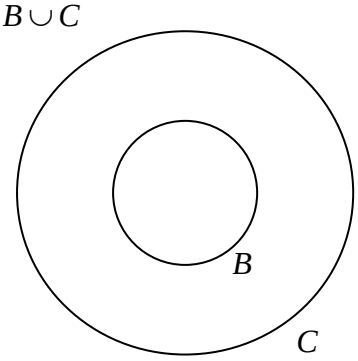
i. If $A = \{1, 3, 5, 7\}$, $B = \{2, 3, 4, 5\}$ and $C = \{1, 2, 3, 4, 5, 6, 7\}$, find

I. $A \cap (B \cup C)$. (2 Marks)

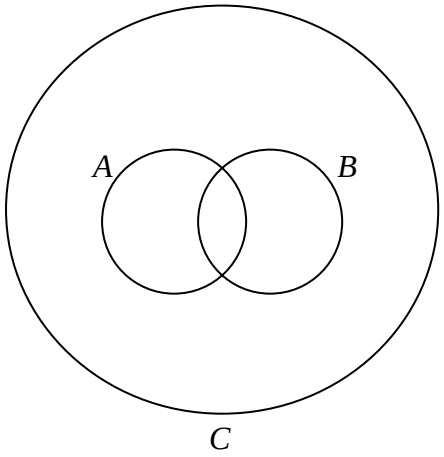
II. $A - (B \cup C)$. (1 Mark)

ii. Shade the following regions in the given Venn diagrams.

I. $B \cup C$ (1 Mark)



II. $A \cap (B \cup C)$ (1 Mark)



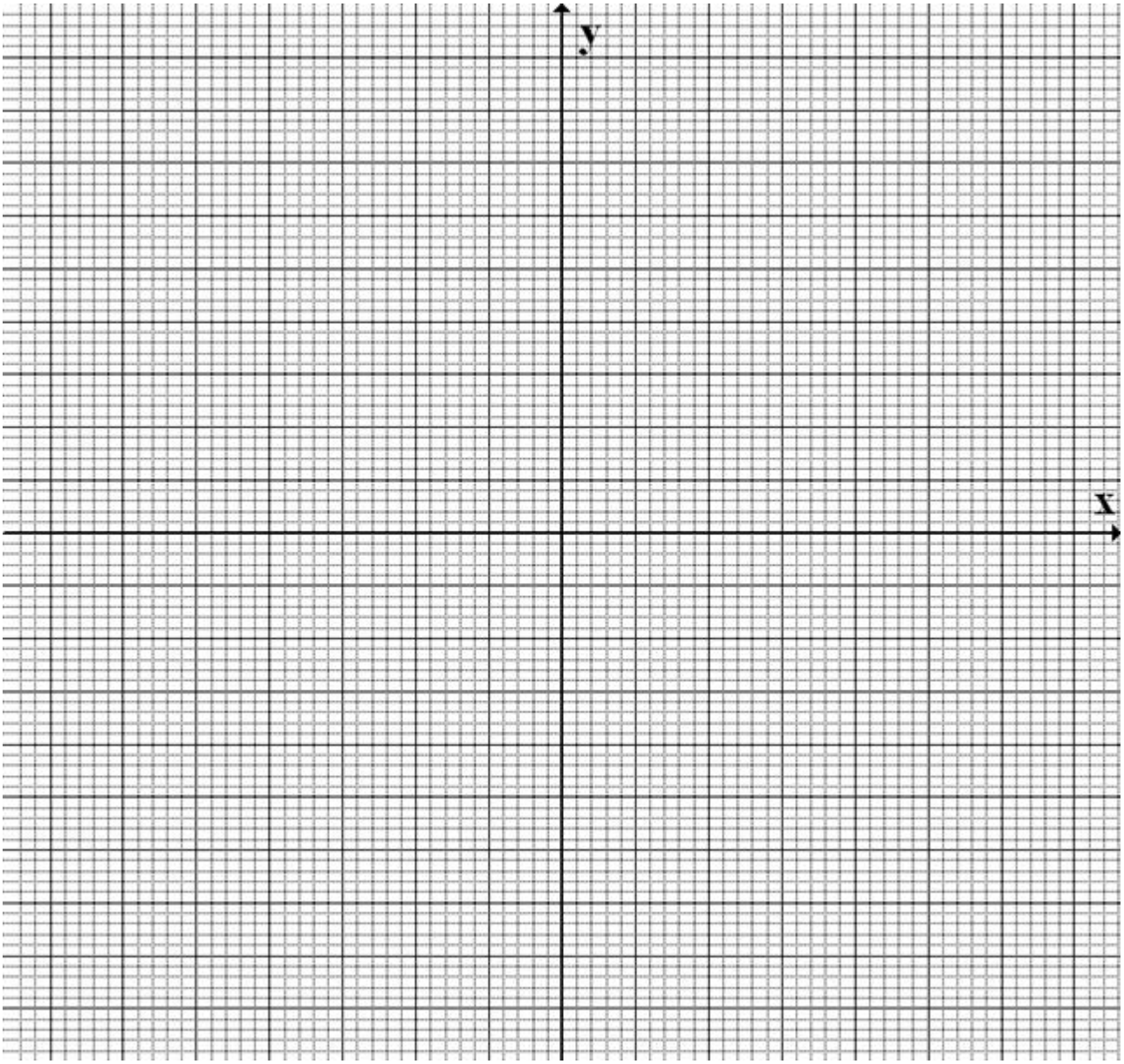
Q.9. (Total 5 Marks)

i. On the given grid, plot the values given in the table below and draw the line $y = 2x - 1$. (2 Marks)

x	-2	0	2
$y = 2x - 1$	-5	-1	3

ii. On the given grid, draw the following lines.

$x = -1$; $y = 2$ (2 Marks)



iii. Write the name of the geometrical figure obtained from the solution. (1 Mark)

(ATTEMPT EITHER PART a OR PART b OF Q.10.)

Q.10. (Total 6 Marks)

a. A small office has a staff of 20 employees. The distance (km) of the office from their homes is shown in the table given below. Complete the table and find the range, variance and standard deviation for this data.

Distance (km)	**Number of Employees**	**Class Mark**		
1 – 4	2	2.5		
5 – 8	1	6.5		
9 – 12	6	10.5		
13 – 16	10	14.5		
17 – 20	1	18.5		
Total		–		

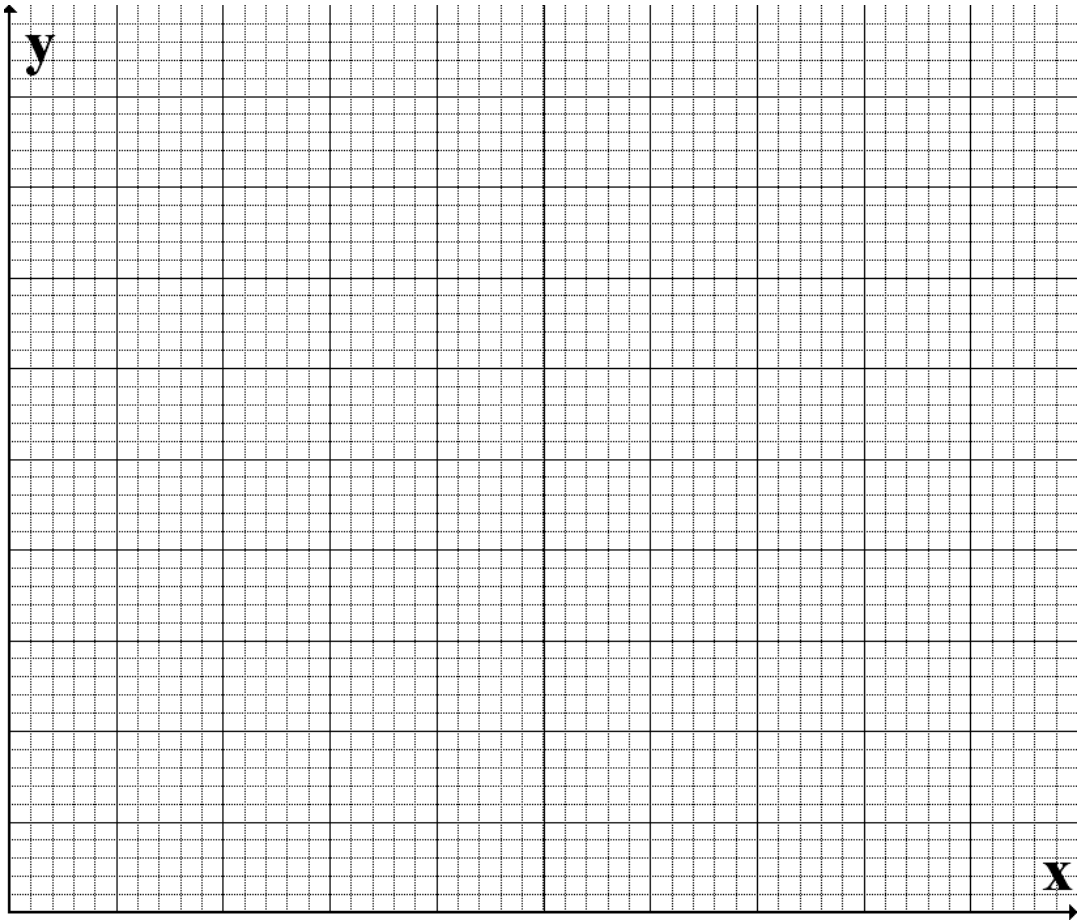
(ATTEMPT EITHER PART a OR PART b OF Q.10.)

b. The given data shows the weight (g) of 60 items.

Weight (g)	**0 – 2**	**2 – 4**	**4 – 6**	**6 – 8**	**8 – 10**	**10 – 12**	**12 – 14**	**14 – 16**	**16 – 18**	**18 – 20**
Frequency	2	5	8	10	12	8	6	5	3	1
Cumulative Frequency	2	7		25	37		51	56		60

i. Complete the given table and use it to construct a cumulative frequency curve. (4 Marks)

ii. Estimate the median weight with the help of the curve. (2 Marks)



END OF PAPER

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